

# Auditable Internal Correspondence Recovery After Logic Synthesis

A proof-carrying artifact for AIG optimization experiments

AIG Optimization Experiments

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## Abstract

This artifact studies when internal correspondences can be recovered after logic synthesis and when they cannot. The central contribution is not a single large recovery count: it is an auditable evidence discipline that separates controlled graph-active successes, blind bounded semantic recovery, exact locality certificates, oracle diagnostics, and historical ineligible rows. The repository includes reproducible checkers, pinned ABC validation, Z3-backed proof paths, and derived paper tables that map every headline claim to concrete CSV rows.

## Problem

Logic synthesis intentionally rewrites internal structure. Names, local topology, support sets, and even apparent semantic signatures can drift under passes such as rewriting, refactoring, resubstitution, and don't-care optimization. This makes internal source mapping and debug correspondence fragile even when whole-design equivalence is easy to prove.

The artifact asks a narrower question: under which evidence conditions can a candidate internal correspondence be counted as recovered, and which blockers explain null results?

## Threat Model

The artifact treats denominators as part of the scientific claim. Controlled generated BLIF rows, blind generated BLIF rows, generated research targets, historical diagnostics, oracle rows, and future external RTL rows are never merged unless a table explicitly says so. A graph rewrite is counted only when the row has graph activity and global equivalence evidence. Diagnostic exact minimum certificates are valuable, but they are not counted as recovered boundaries unless a validated graph artifact exists.

## Method

The repository implements four connected proof paths.

1. Source-side counterpart construction selects graph-active source expressions and proves local equivalence before synthesizing a quotient.
2. Cross-netlist transplantation builds input and output adapters and requires local proof plus whole-design CEC before accepting a transplant.
3. Blind semantic CEGIS enumerates bounded template candidates and records concrete counterexamples whenever a candidate is refuted.
4. Necessity-first and formal locality audits explain why historical failures remain failures: missing provenance, target irrelevance, non-compact exact interfaces, or absent validated rewrite artifacts.

# Main Result

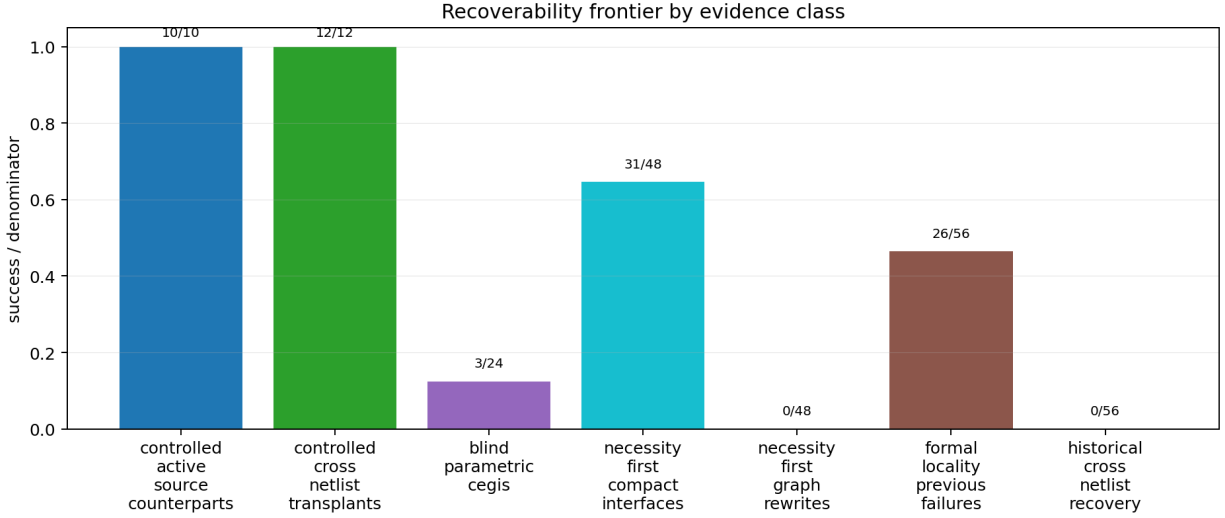


Figure 1: Recoverability frontier by evidence class

The main figure is generated by `make research-wow` from committed tables in `results/`. It intentionally places controlled successes next to blind, generated, formal-diagnostic, and historical denominators so the evidence level is visible in the plot itself.

## Evidence Summary

The canonical paper-facing table is `results/research Wow/recoverability_frontier.csv`. It records:

- 10/10 controlled graph-active source-counterpart rewrites.
- 12/12 controlled cross-netlist transplants.
- 3/24 blind parametric CEGIS regions formally verified.
- 31/48 necessity-first generated targets with compact exact input interfaces.
- 0/48 necessity-first generated targets with validated graph rewrite artifacts.
- 26/56 historical formal-locality diagnostic rows with exact input minima.
- 0/56 corrected historical cross-netlist rows counted as graph-active recoveries.

## Failure Taxonomy

The failure taxonomy in `results/research Wow/failure_taxonomy.csv` converts null results into auditable blocker classes. The largest classes are missing optimized artifacts, target irrelevance after reconstruction, non-compact exact interfaces, and absent validated rewrite artifacts. These categories prevent a reviewer from mistaking historical or diagnostic rows for failed executions of the final graph-rewrite contract.

## Demo Case Study

`make demo-wow` prints a reviewer-safe trace that combines a controlled positive proof path with a blind counterexample-refinement trace. The controlled case proves a graph-active source counterpart, decomposes the source window, synthesizes an exact quotient, and checks source/cross equivalence with ABC. The blind case shows two concrete SAT counterexamples before bounded candidate exhaustion, plus a separate blind positive control with an exhaustive formal proof.

## Ablations and Baselines

The derived ablation and baseline summaries in `results/research_wow/` show that successful recovery depends on the proof-carrying construction and on strict provenance filtering. Baselines that produce local or diagnostic evidence are kept separate from graph-active recovery counts.

## Reproduction

Reviewer-safe validation:

```
python -m pip install -r requirements.txt
make artifact-check
make paper-pdf
```

Full formal validation with pinned ABC:

```
make formal-abc
```

The PDF is written to `output/pdf/aig_internal_correspondence_artifact.pdf`.

## Limitations

External RTL claims remain out of scope until a pinned redistributable RTL corpus and Yosys lowering flow are committed. Oracle rows are diagnostic only. Approximate or sampled evidence remains approximate or sampled evidence. Exact locality certificates do not imply graph-active rewrites without a validated artifact and whole-design CEC.

## Related Work Positioning

The work is adjacent to equivalence checking, SAT sweeping and FRAIG, internal signal correspondence, observability don't-care reasoning, source mapping, and formal artifact evaluation. The artifact's main distinction is its claim discipline: each result is labeled by evidence level, denominator class, and checker command, so positive and negative results are both reproducible.